Advanced ML Case Study: Fraudulent Claim Detection

Questions

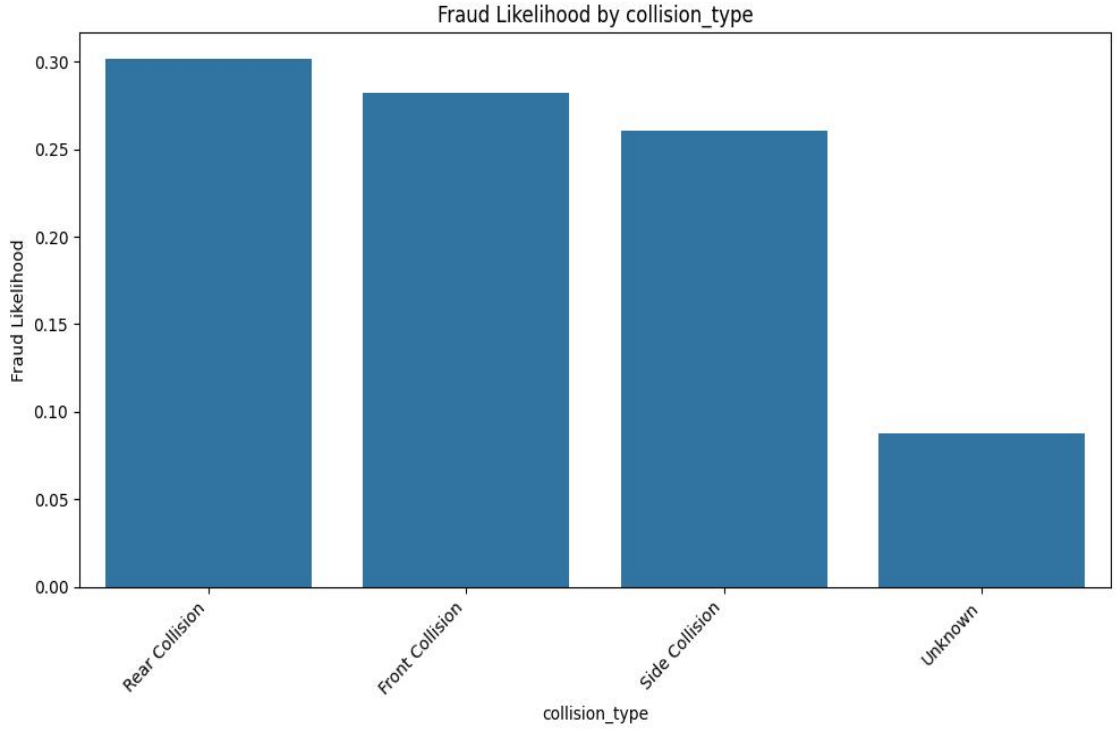
1. How can we analyse historical claim data to detect patterns that indicate fraudulent claims?
2. Which features are most predictive of fraudulent behaviour?
3. Can we predict the likelihood of fraud for an incoming claim, based on past data?
4. What insights can be drawn from the model that can help in improving the fraud detection process?

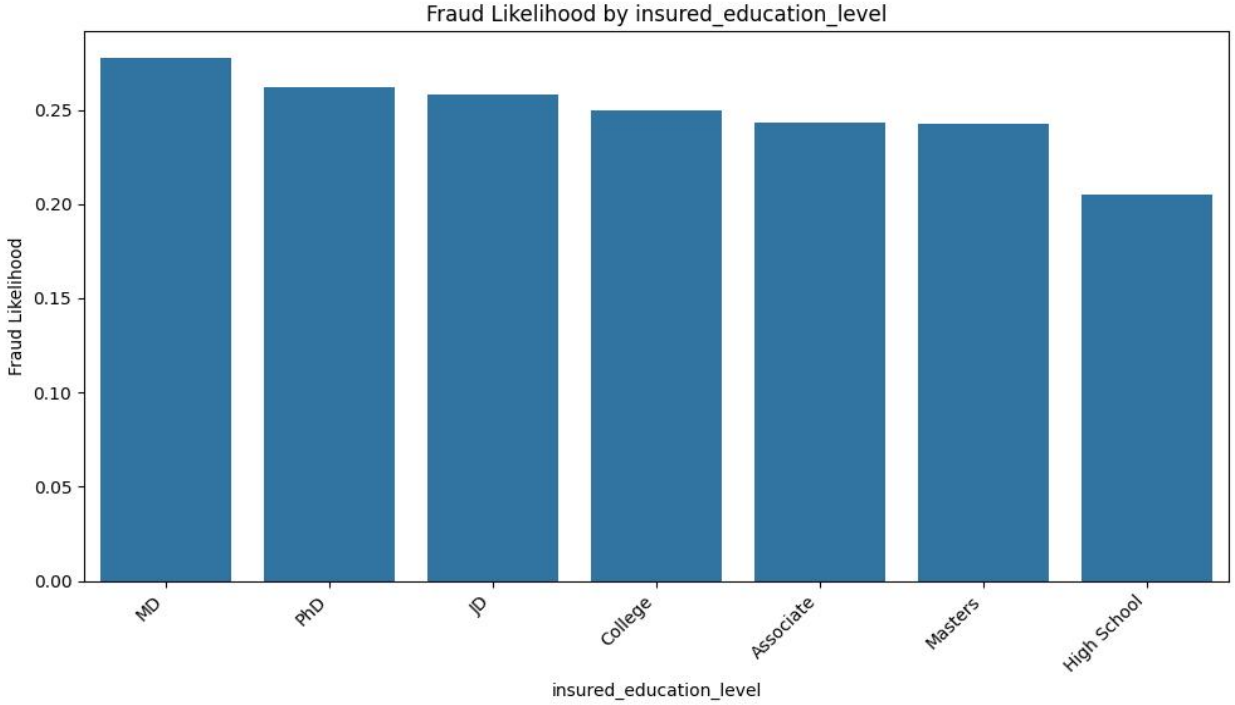
1. How can we analyse historical claim data to detect patterns that indicate fraudulent claims?

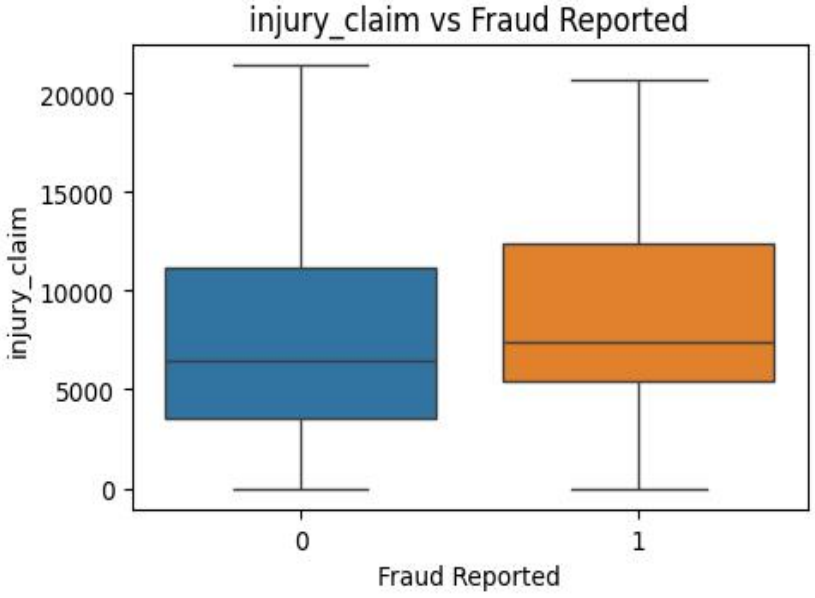
To identify patterns indicative of fraudulent claims, we analyze historical data by examining how various features correlate with the likelihood of a claim being fraudulent. This involves two primary approaches:

- **Comparative Bar Plots for Categorical Features**: For categorical features, we visualize the proportion of fraudulent claims within each category. A significant variation in fraud likelihood across different categories of a feature suggests that this feature is a strong indicator of fraud. For example, if a specific incident_type has a much higher fraud rate than others, it highlights a potential pattern.

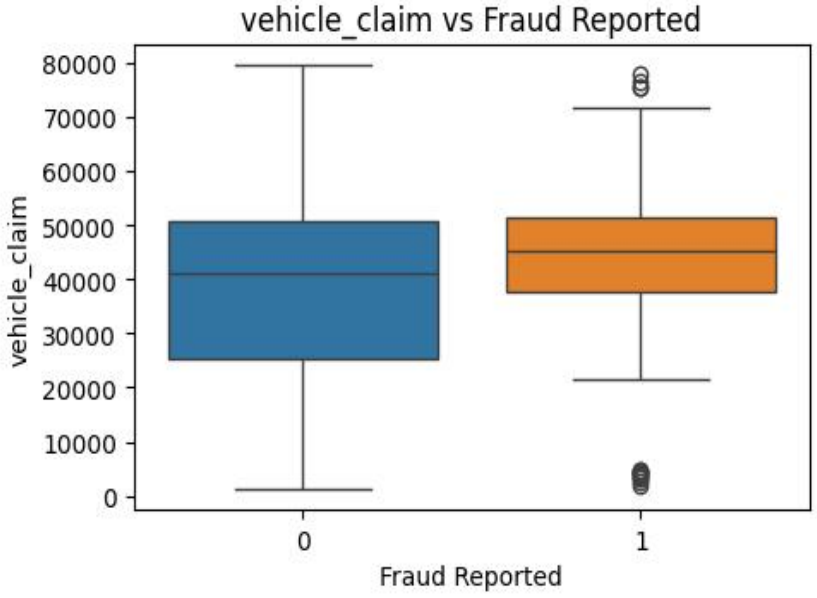
- **Box Plots for Numerical Features**: For numerical features, we compare their distributions for legitimate versus fraudulent claims. Differences in the median, spread, and presence of outliers between these two groups can indicate that the numerical feature plays a role in differentiating fraudulent claims. For instance, if total_claim_amount tends to be higher or lower for fraudulent claims, this provides valuable insight.


Fraud Likelihood by collision_type
Fraud Likelihood
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Rear Collision
Front Collision
Side Collision
Unknown
collision_type


Fraud Likelihood by insured_education_level
Fraud Likelihood
0.25
0.20
0.15
0.10
0.05
0.00
MD
PhD
JD
College
Associate
Masters
High School
insured_education_level

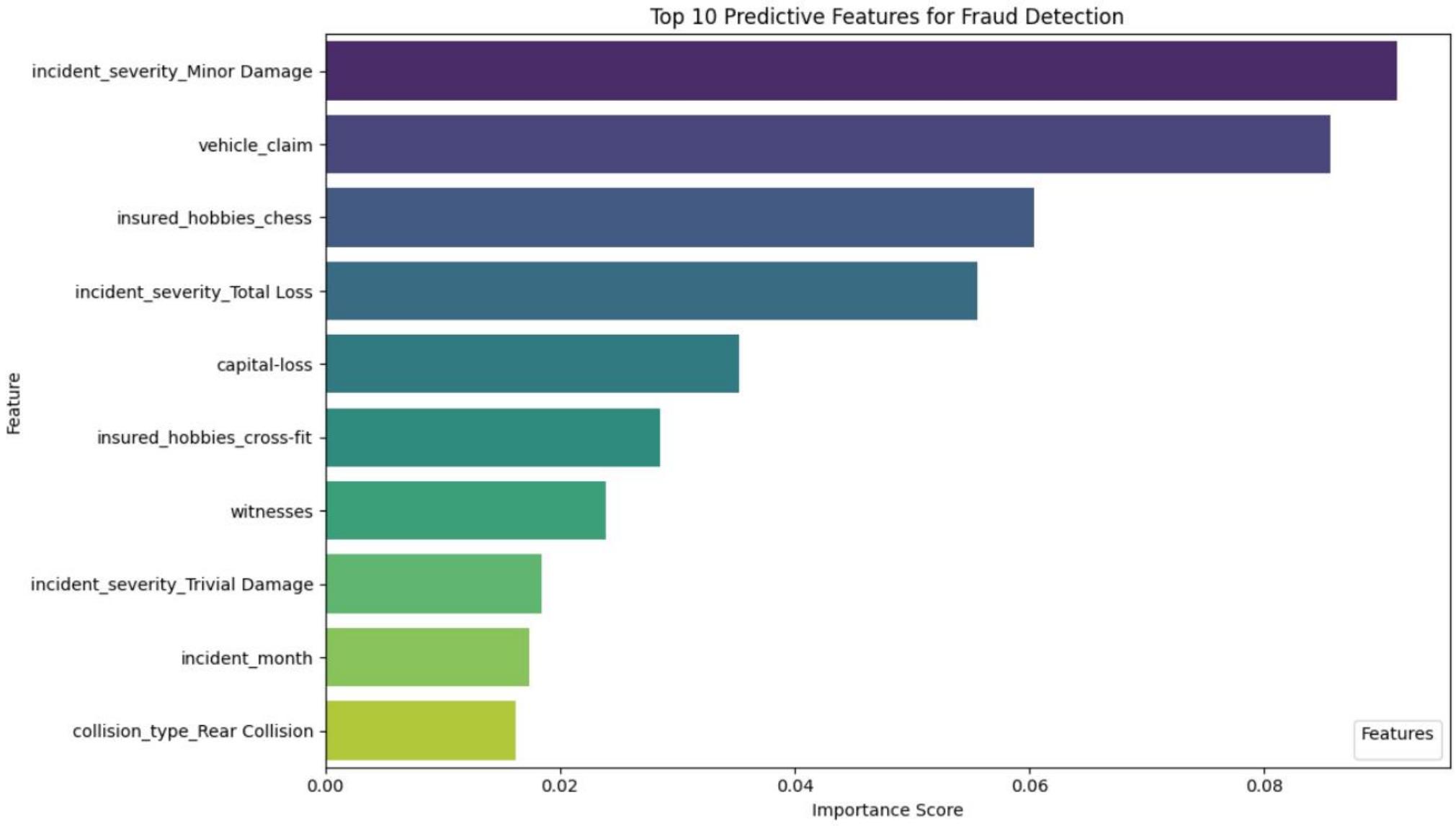

injury_claim vs Fraud Reported
injury_claim
20000
15000
10000
5000
0
0
1
Fraud Reported

property_claim vs Fraud Reported
property_claim
20000
15000
10000
5000
0
0
1
Fraud Reported


vehicle_claim vs Fraud Reported
vehicle_claim
80000
70000
60000
50000
40000
30000
20000
10000
0
0
1
Fraud Reported

2. Which features are most predictive of fraudulent behaviour?

- Feature importance scores, derived from the Random Forest model, quantify the contribution of each feature to the model's predictive accuracy. These scores indicate how much each feature decreases the impurity (or Gini impurity in the case of decision trees within a Random Forest) when splitting nodes in the trees.
- A **higher importance score** for a feature suggests that it has a stronger influence on the model's ability to classify insurance claims as fraudulent or legitimate. In simpler terms, features with higher scores are more predictive of fraudulent behavior. By examining these scores, we can pinpoint which aspects of a claim (e.g., incident details, customer profile, claim amounts) are most crucial for identifying fraud.
- This analysis helps us understand the underlying drivers of fraud and can guide further investigations or inform business strategies to prevent future fraudulent activities.


Top 10 Predictive Features for Fraud Detection
incident_severity_Minor Damage
vehicle_claim
insured_hobbies_chess
incident_severity_Total Loss
capital-loss
insured_hobbies_cross-fit
witnesses
incident_severity_Trivial Damage
incident_month
collision_type_Rear Collision
Feature
Features
0.00
0.02
0.04
0.06
0.08
Importance Score

3. Can we predict the likelihood of fraud for an incoming claim, based on past data?

To predict the likelihood of fraud for incoming claims, we utilize the **Tuned Random Forest model**, which has been optimized and validated on historical data.

When a new claim is submitted, its features are first preprocessed in the same manner as the training data, including scaling numerical features and one-hot encoding categorical features, and selecting only the most important features identified during model building.

The preprocessed new claim data is then fed into the rf_tuned_model. The model outputs a probability score, indicating the likelihood that the claim is fraudulent. A higher probability score suggests a greater chance of fraud.

The Tuned Random Forest model demonstrated the following performance metrics on the validation dataset:

Accuracy: 82.00% **F1-Score**: 0.675 **Sensitivity (Recall)**: 75.68%

Specificity: 84.07% **Precision**: 60.87%

These metrics collectively indicate that the model is reasonably effective at identifying fraudulent claims, with a good balance between correctly identifying positive cases (fraud) and correctly identifying negative cases (legitimate claims).

4. What insights can be drawn from the model that can help in improving the fraud detection process?

Based on the comprehensive analysis and model building, several actionable insights and recommendations emerge to significantly enhance Global Insure's fraud detection capabilities:

Deploy the Tuned Random Forest Model: The Tuned Random Forest model demonstrated superior performance on the validation data (Accuracy: 82.00%, F1-Score: 0.675) compared to the Logistic Regression model. Its ability to handle complex non-linear relationships and provide robust predictions makes it the most suitable candidate for real-world deployment.

- Implementing this model early in the claims approval process will enable data-driven classification of claims as fraudulent or legitimate, minimizing financial losses associated with undetected fraud.

Focus on Key Predictive Features: The Random Forest model identified specific features as highly predictive of fraudulent behavior. These include:

- incident_severity_Minor Damage and incident_severity_Total Loss: The severity of the incident plays a crucial role. Claims with minor or total damage seem to have different fraud patterns.
- vehicle_claim: The amount claimed for vehicle damage is a significant indicator.
- insured_hobbies_chess and insured_hobbies_cross-fit: Surprisingly, certain insured hobbies appear to correlate with fraud likelihood, suggesting potential demographic or behavioral patterns among fraudsters.
- capital-loss: Capital losses might indicate financial distress, which could be a motivator for fraudulent claims.
- witnesses: The number of witnesses present at the scene can influence the claim's legitimacy.

Global Insure should leverage these insights by:

- **Refining Data Collection**: Ensure accurate and detailed recording of these key features for all incoming claims.
- **Targeted Investigations**: Claims exhibiting suspicious values or combinations within these features should be prioritized for deeper investigation by human analysts.
- **Training and Awareness**: Educate claims adjusters and investigative teams on these specific indicators to improve their manual assessment skills.

Implement Continuous Monitoring for Model Performance: Machine learning models are not static; their performance can degrade over time due to shifts in data patterns, known as concept drift. Therefore, it is critical to:

- **Regularly Evaluate**: Continuously monitor the deployed model's accuracy, precision, recall, and F1-score on new, incoming data.
- **Feedback Loops**: Establish a feedback mechanism where the outcomes of claims investigated (e.g., confirmed fraud or legitimate) are fed back into the system to retrain and refine the model periodically. This ensures the model remains effective against evolving fraud tactics.
- **Alert Systems**: Set up alerts for significant drops in model performance metrics to prompt immediate review and potential retraining.

Explore Advanced Machine Learning Techniques in the Future: While the Random Forest model is a strong start, further improvements can be explored by investigating more sophisticated techniques:

- **Gradient Boosting Models (e.g., XGBoost, LightGBM)**: These models often achieve state-of-the-art performance in tabular data classification tasks and could potentially yield even higher accuracy and F1-scores.
- **Anomaly Detection**: Techniques like Isolation Forests or One-Class SVMs could be explored to identify claims that deviate significantly from historical legitimate patterns, potentially uncovering novel fraud schemes.
- **Deep Learning (e.g., Neural Networks)**: For very large and complex datasets, deep learning models might uncover intricate patterns that traditional ML models might miss.

- By systematically implementing these recommendations, Global Insure can establish a robust, adaptive, and highly effective fraud detection system, significantly reducing financial losses and optimizing operational efficiency.

Thank you

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